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NAAC ACCREDITED WITH A++ GRADE

WOMEN SAFETY · IOT PROJECT

# SafeGuard +

SOS

Smart IoT Wearable for Women with GPS and SOS Alert

## Introduction

SafeGuard + is a small wearable device that sends the wearer's live GPS location to trusted contacts by SMS with one button press, or automatically when a fall is detected.

## Problem Statement

Women often face unsafe situations where there is no time to unlock a phone, call for help or share a location. A quick, always-on and low-cost alert device is needed.

## Objective

- One-press SOS alert from a wearable.
- Send GPS location to contacts without internet.
- Detect falls and alert automatically.

## Work Flow

**1 Trigger** — Button press or fall detected

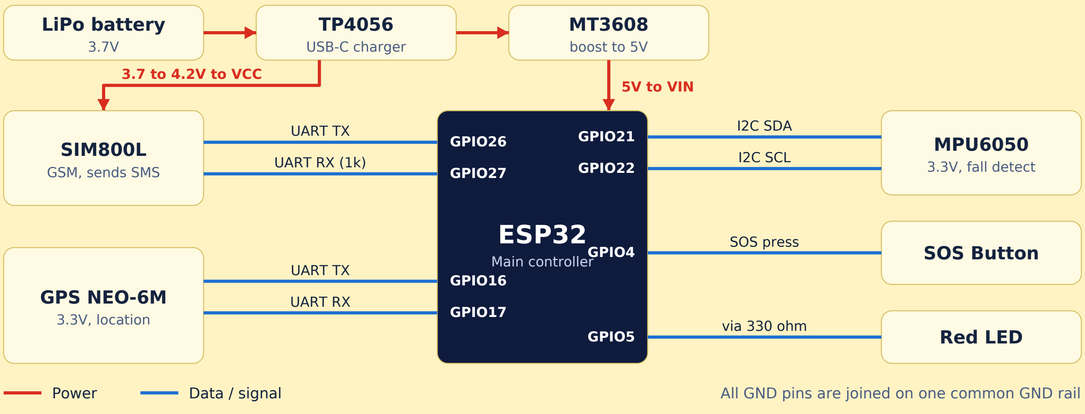
**2 ESP32** — Processes the alert

**3 GPS** — Reads latitude and longitude

**4 SIM800L** — Sends SMS with map link

**5 Contacts** — Family gets the alert

## System Structure (Circuit Connections)




## Key Features

**01 One-press SOS**
Instant alert to saved contacts

**02 Live GPS location**
Google Maps link in the SMS

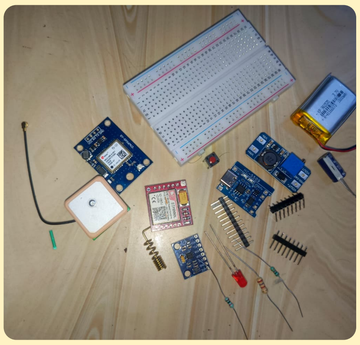
**03 Fall detection**
MPU6050 with 10 second cancel

**04 Works without internet**
SMS over the GSM network

**05 Rechargeable, wearable**
LiPo battery, USB-C charging

**06 Low cost**
ESP32 and Arduino code

## Devices Used








ESP32 · NEO-6M GPS · SIM800L · MPU6050 · TP4056 · MT3608 · LiPo battery

## Code Snippet

```
if (digitalRead(BUTTON_PIN) == LOW) {
  String msg = "SOS! Madad chahiye. ";
  msg += "maps.google.com/?q=";
  msg += String(gps.location.lat(), 6);
  msg += ",";
  msg += String(gps.location.lng(), 6);
  sendSMS(NUMBERS[0], msg);
  blinkLed(3, 300);
}
```

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